

DuPont™ Delrin® 500P NC010

ACETAL RESIN

Product Information

Common features of Delrin® acetal resins include mechanical and physical properties such as high mechanical strength and rigidity, excellent fatigue and impact resistance, as well as resistance to moisture, gasoline, lubricants, solvents, and many other neutral chemicals. Delrin® acetal resins also have excellent dimensional stability and good electrical insulating characteristics. They are naturally resilient, self-lubricating, and available in a variety of colors and specialty grades.

Delrin® acetal resin typically is used in demanding applications in the automotive, domestic appliances, sports, industrial engineering, electronics, and consumer goods industries.

Delrin® 500P is a general purpose medium viscosity acetal homopolymer for injection molding. It has improved processing thermal stability, a good combination of mechanical properties, and low VOC emissions.

General information	Value	Unit	Test Standard
Resin Identification	POM	-	ISO 1043
Part Marking Code	POM	-	ISO 11469
Rheological properties	Value	Unit	Test Standard
Melt volume-flow rate	13	cm ³ /10min	ISO 1133
Melt mass-flow rate	15	g/10min	ISO 1133
Temperature	190	°C	ISO 1133
Load	2.16	kg	ISO 1133
Melt mass-flow rate, Temperature	190	°C	ISO 1133
Melt mass-flow rate, Load	2.16	kg	ISO 1133
Moulding shrinkage, parallel	2.0	%	ISO 294-4, 2577
Moulding shrinkage, normal	1.9	%	ISO 294-4, 2577
Mechanical properties	Value	Unit	Test Standard
Tensile Modulus	3100	MPa	ISO 527-1/-2
Yield stress	71	MPa	ISO 527-1/-2
Yield strain	17	%	ISO 527-1/-2
Nominal strain at break	30	%	ISO 527-1/-2
Flexural Modulus	3000	MPa	ISO 178
Flexural Stress at 3.5%	80	MPa	ISO 178
Tensile creep modulus			ISO 899-1
1h	2800	MPa	
1000h	1600	MPa	
Charpy impact strength			ISO 179/1eU
23°C	320	kJ/m ²	
-30°C	280	kJ/m ²	
Charpy notched impact strength			ISO 179/1eA
23°C	9	kJ/m ²	
-30°C	8	kJ/m ²	
Puncture - maximum force, 23°C	2000	N	ISO 6603-2
Puncture energy, 23°C	3	J	ISO 6603-2
Izod notched impact strength			ISO 180/1A
23°C	9	kJ/m ²	
-30°C	8	kJ/m ²	
Izod impact strength			ISO 180/1U
23°C	280	kJ/m ²	
-30°C	250	kJ/m ²	
Hardness, Rockwell, M-scale	92	-	ISO 2039-2
Hardness, Rockwell, R-scale	120	-	ISO 2039-2
Ball indentation hardness, H 358/30	192	MPa	ISO 2039-1
Ball indentation hardness, H 961/30	170	MPa	ISO 2039-1
Poisson's ratio	0.37	-	-
Thermal properties	Value	Unit	Test Standard
Melting temperature, 10°C/min	178	°C	ISO 11357-1/-3

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Temp. of deflection under load			ISO 75-1/-2
1.8 MPa	95	°C	
1.8 MPa, annealed	110	°C	
0.45 MPa	160	°C	
Vicat softening temperature, 50°C/h, 50N	155	°C	ISO 306
Ball pressure test	165	°C	IEC 60695-10-2
Coeff. of linear therm. expansion, parallel	100	E-6/K	ISO 11359-1/-2
Coeff. of linear therm. expansion			ISO 11359-1/-2
normal	100	E-6/K	
Normal, -40-23°C	90	E-6/K	
Parallel, -40-23°C	90	E-6/K	
Thermal conductivity of melt	0.24	W/(m K)	-
Eff. thermal diffusivity	9E-8	m²/s	-
RTI, electrical			UL 746B
0.75mm	50	°C	
1.5mm	110	°C	
3mm	110	°C	
RTI, impact			UL 746B
0.75mm	50	°C	
1.5mm	85	°C	
3mm	90	°C	
RTI, strength			UL 746B
0.75mm	50	°C	
1.5mm	90	°C	
3mm	95	°C	
Flammability			
Burning Behav. at 1.5mm nom. thickn.	HB	class	IEC 60695-11-10
Thickness tested	1.5	mm	IEC 60695-11-10
UL recognition	yes	-	UL 94
Burning Behav. at thickness h	HB	class	IEC 60695-11-10
Thickness tested	0.8	mm	IEC 60695-11-10
UL recognition	yes	-	UL 94
Oxygen index	22	%	ISO 4589-1/-2
Glow Wire Flammability Index			IEC 60695-2-12
1mm	550	°C	
2mm	550	°C	
3mm	550	°C	
FMVSS Class	B	-	ISO 3795 (FMVSS 302)
Burning rate, Thickness 1 mm	20	mm/min	ISO 3795 (FMVSS 302)
Electrical properties			
Relative permittivity			IEC 62631-2-1
100Hz	3.8	-	
1MHz	3.8	-	
Dissipation factor			IEC 62631-2-1
100Hz	90	E-4	
1MHz	55	E-4	
Volume resistivity	2E12	Ohm*m	IEC 62631-3-1
Surface resistivity	4E14	Ohm	IEC 62631-3-2
Electric strength	44	kV/mm	IEC 60243-1
Comparative tracking index	600	-	IEC 60112
Other properties			
Humidity absorption, 2mm	0.2	%	Sim. to ISO 62
Water absorption, 2mm	1.3	%	Sim. to ISO 62
Density	1420	kg/m³	ISO 1183
Density of melt	1190	kg/m³	-

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VDA Properties	Value	Unit	Test Standard
Emissions		mg/kg	VDA 275
Fogging, F-value (refraction)	90	%	ISO 6452
Fogging, G-value (condensate)	0.35	mg	ISO 6452

1:

Injection	Value	Unit	Test Standard
Drying Recommended	yes		-
Drying Temperature	≥80	°C	-
Drying Time, Dehumidified Dryer	2 - 4	h	-
Processing Moisture Content	≤0.2	%	-
Melt Temperature Optimum	215	°C	-
Min. melt temperature	210	°C	-
Max. melt temperature	220	°C	-
Mold Temperature Optimum	90	°C	-
Min. mould temperature	80	°C	-
Max. mould temperature	100	°C	-
Hold pressure range	80 - 100	MPa	-
Hold pressure time	8	s/mm	-
Annealing time, optional	30	min/mm	-
Annealing temperature	160	°C	-

Characteristics

Processing	• Injection Moulding		
Delivery form	• Pellets		
Additives	• Lubricants	• Release agent	
Regional Availability	• North America • Europe	• Asia Pacific • South and Central America	• Near East/Africa • Global

Processing Texts

Injection molding

Drying is recommended, but not necessary for newly opened packaging stored in a dry location.

Follow the drying guidelines above in the following cases:

- If moisture is above the Processing Moisture Content recommendation,
- When a resin container is damaged,
- When the material is not properly stored in a dry place at room temperature, or
- When packaging stays open for a significant time.

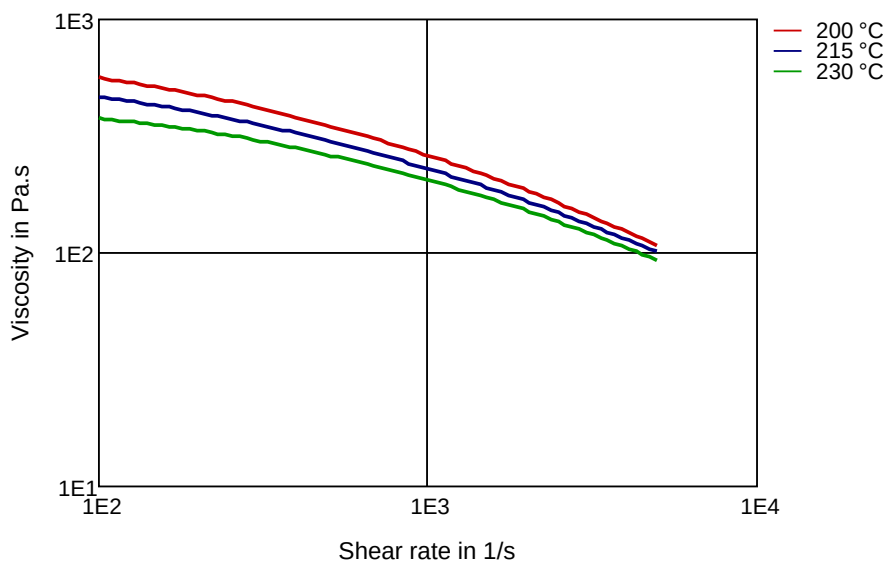


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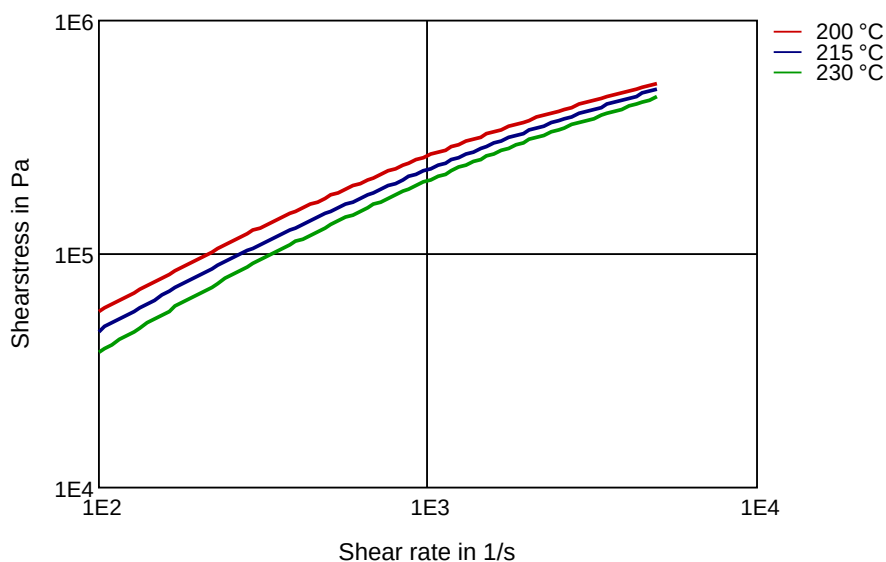
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Diagrams

Viscosity-shear rate



Shearstress-shear rate



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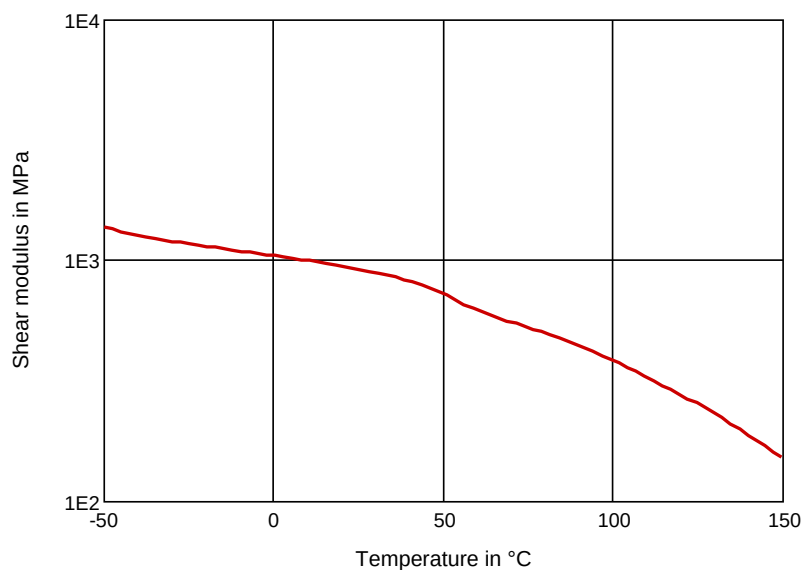
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Dynamic Shear modulus-temperature



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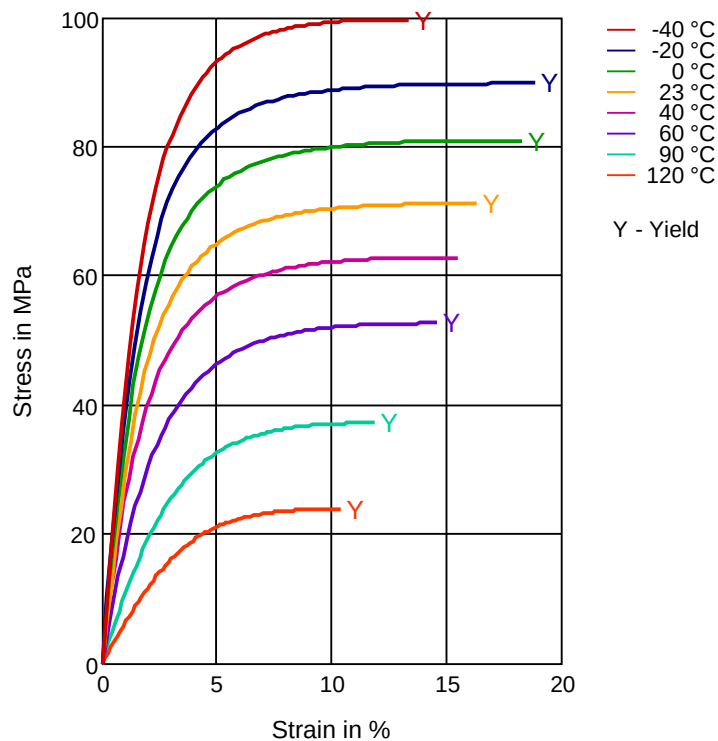
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Stress-strain



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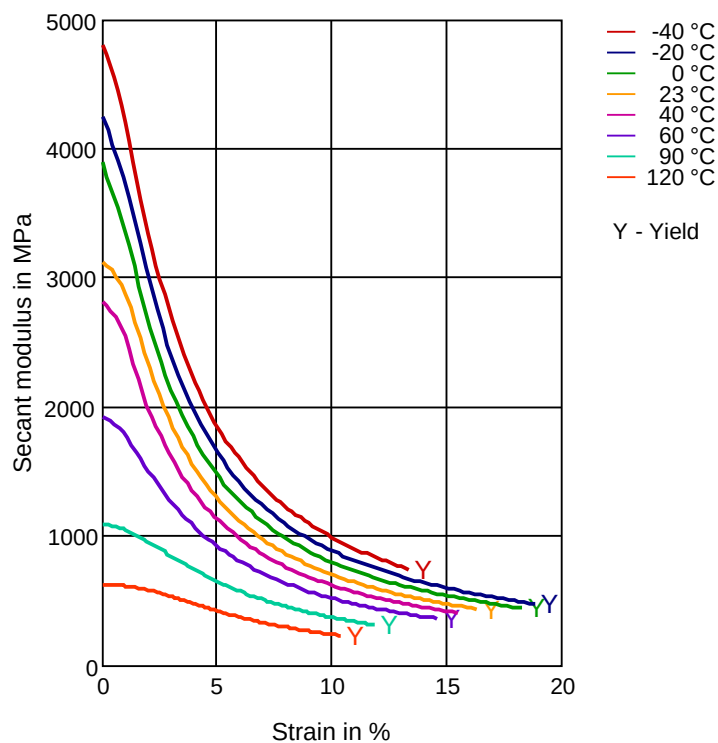


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Secant modulus-strain



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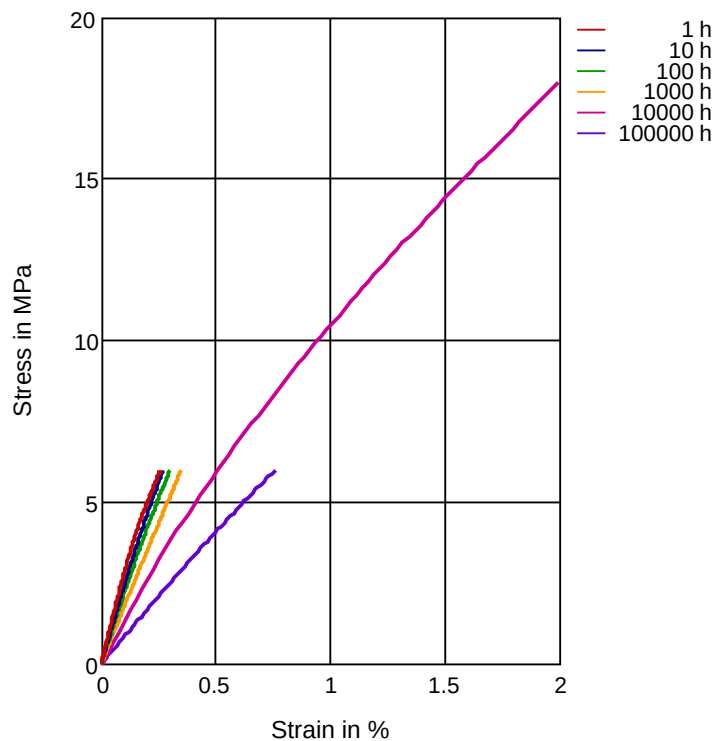


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Stress-strain (isochronous) 23 °C



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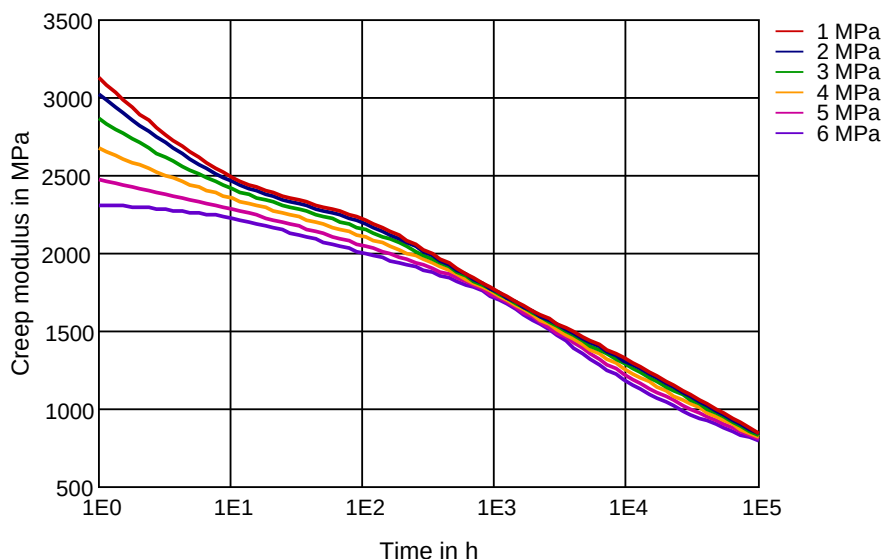
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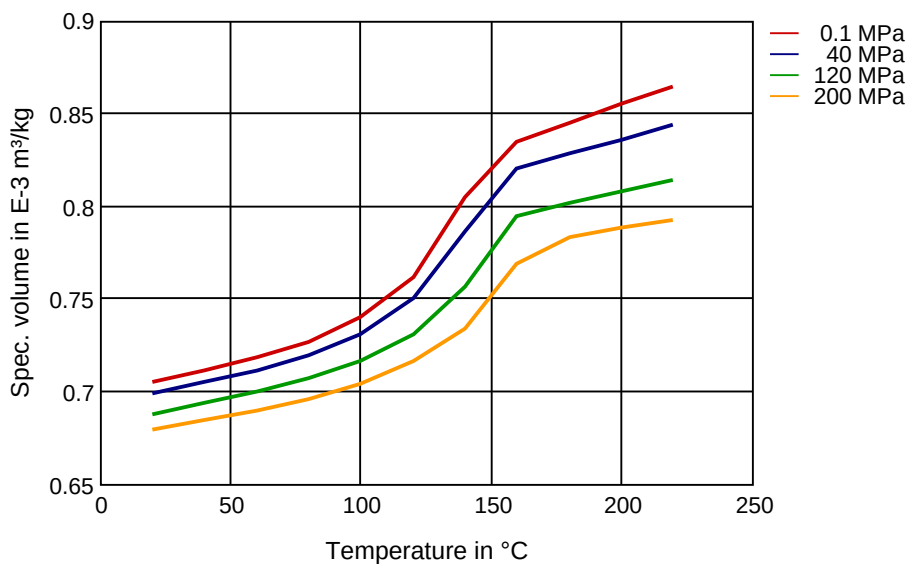
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Creep modulus-time 23 °C



Specific volume-temperature (pvT)



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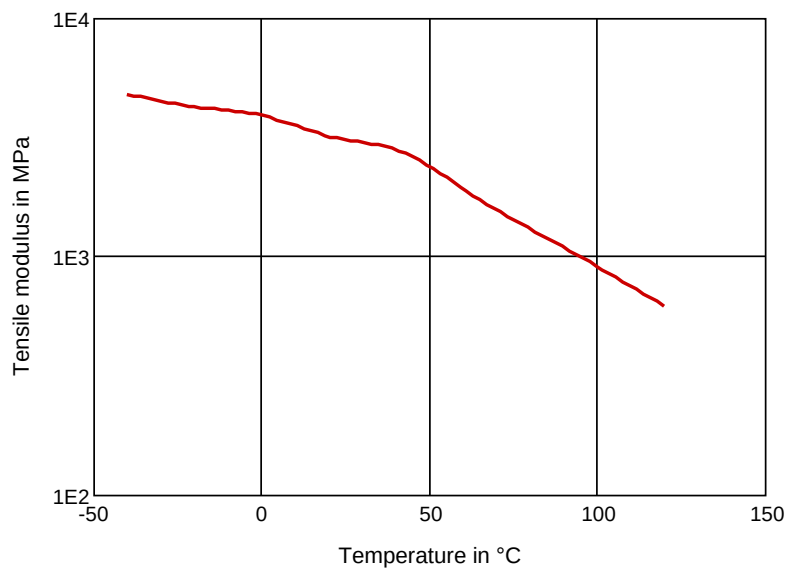
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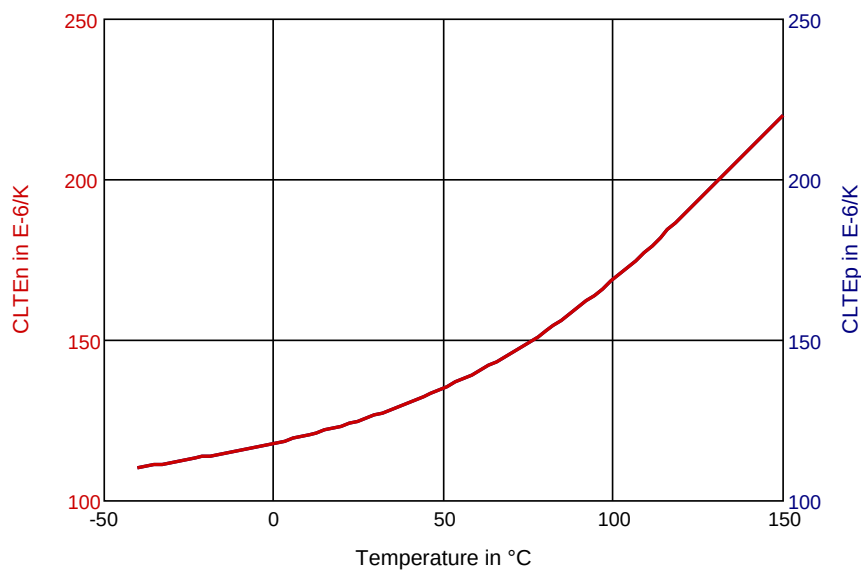
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Tensile modulus-temperature



Coeff. of linear thermal expansion, normal



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Chemical Media Resistance

Acids

- ✓ Acetic Acid (5% by mass) (23°C)
- ✗ Citric Acid solution (10% by mass) (23°C)
- ✗ Lactic Acid (10% by mass) (23°C)
- ✗ Hydrochloric Acid (36% by mass) (23°C)
- ✗ Nitric Acid (40% by mass) (23°C)
- ✗ Sulfuric Acid (38% by mass) (23°C)
- ✗ Sulfuric Acid (5% by mass) (23°C)
- ✗ Chromic Acid solution (40% by mass) (23°C)

Bases

- ✗ Sodium Hydroxide solution (35% by mass) (23°C)
- ✗ Sodium Hydroxide solution (1% by mass) (23°C)
- ✗ Ammonium Hydroxide solution (10% by mass) (23°C)

Alcohols

- ✓ Isopropyl alcohol (23°C)
- ✓ Methanol (23°C)
- ✓ Ethanol (23°C)

Hydrocarbons

- ✓ n-Hexane (23°C)
- ✓ Toluene (23°C)
- ✓ iso-Octane (23°C)

Ketones

- ✓ Acetone (23°C)

Ethers

- ✓ Diethyl ether (23°C)

Mineral oils

- ✓ SAE 10W40 multigrade motor oil (23°C)
- ✗ SAE 10W40 multigrade motor oil (130°C)
- ✗ SAE 80/90 hypoid-gear oil (130°C)
- ✓ Insulating Oil (23°C)

Standard Fuels

- ✓ ISO 1817 Liquid 1 - E5 (60°C)
- ✓ ISO 1817 Liquid 2 - M15E4 (60°C)
- ✓ ISO 1817 Liquid 3 - M3E7 (60°C)
- ✓ ISO 1817 Liquid 4 - M15 (60°C)
- ✓ Standard fuel without alcohol (pref. ISO 1817 Liquid C) (23°C)
- ✓ Standard fuel with alcohol (pref. ISO 1817 Liquid 4) (23°C)



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- ✓ Diesel fuel (pref. ISO 1817 Liquid F) (23°C)
- ✗ Diesel fuel (pref. ISO 1817 Liquid F) (90°C)
- ✗ Diesel fuel (pref. ISO 1817 Liquid F) (>90°C)
- ✗ Diesel EN 590 (100°C)

Salt solutions

- ✓ Sodium Chloride solution (10% by mass) (23°C)
- ✗ Sodium Hypochlorite solution (10% by mass) (23°C)
- ✗ Sodium Carbonate solution (20% by mass) (23°C)
- ✗ Sodium Carbonate solution (2% by mass) (23°C)
- ✗ Zinc Chloride solution (50% by mass) (23°C)

Other

- ✓ Ethyl Acetate (23°C)
- ✗ Hydrogen peroxide (23°C)
- ✗ DOT No. 4 Brake fluid (130°C)
- ✗ Ethylene Glycol (50% by mass) in water (108°C)
- ✓ 1% nonylphenoxy-polyethyleneoxy ethanol in water (23°C)
- ✓ 50% Oleic acid + 50% Olive Oil (23°C)
- ✓ Water (23°C)
- ✗ Water (90°C)
- ✗ Phenol solution (5% by mass) (23°C)

Symbols used:

✓ possibly resistant

Defined as: Supplier has sufficient indication that contact with chemical can be potentially accepted under the intended use conditions and expected service life. Criteria for assessment have to be indicated (e.g. surface aspect, volume change, property change).

✗ not recommended - see explanation

Defined as: Not recommended for general use. However, short-term exposure under certain restricted conditions could be acceptable (e.g. fast cleaning with thorough rinsing, spills, wiping, vapor exposure).

Contact DuPont for Material Safety Data Sheet, general guides and/or additional information about ventilation, handling, purging, drying, etc. ISO Mechanical properties measured at 4mm (Hytrel® measured at 2 mm), IEC Electrical properties measured at 2mm, all ASTM properties measured at 3.2mm, and test temperatures are 23°C unless otherwise stated.

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